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Online detection and model-free adaptive control of titanium alloy laser welding penetration depth based on spectral diagnosis

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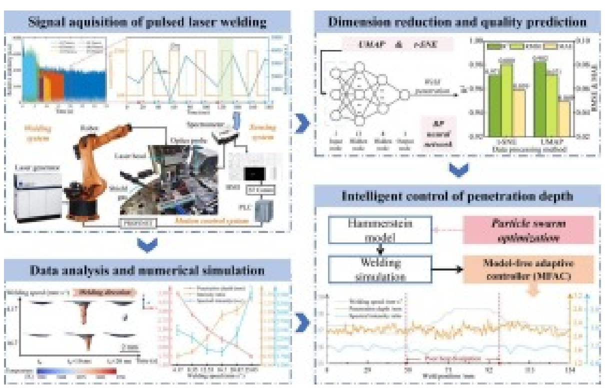
Outline

Abstract

The stable weld penetration depth is of great significance for ensuring the joint strength. However, due to the complex welding heat dissipation conditions, online detection and control of laser welding penetration depth remains a great challenge. This study analyzed the variation mechanism of laser-induced plasma spectrum and its relationship with penetration depth in pulsed laser welding. To acquire valuable interaction information between the laser and the base metal from the redundant spectral data, several feature extraction were investigated. A multi-layer backpropagation neural network was introduced to predict penetration depth. The results showed that the model based on spectral features acquired by uniform manifold approximation and projection (UMAP) had the highest accuracy in predicting the penetration depth, with R^2 of 0.982. Since the prediction period of data processed by UMAP was longer than that of spectral intensity ratio of Ti I 503.995 nm / Ti I 586.919 nm based on the spectral emission theory, the spectral intensity ratio was selected as the controlled object to ensure the real-time performance of penetration depth control. The nonlinear characteristics of the pulsed laser welding process were identified through step response experiments. Furthermore, an optimized Hammerstein nonlinear system model was constructed to verify that the spectral intensity ratio accurately characterized the penetration depth. In this paper, a model-free adaptive control algorithm was adopted to develop a control system of penetration depth. The simulation results showed that the intelligent controller had a short regulation time and a small overshoot. The welding test on dumbbell-shaped workpieces showed that the designed control system well achieved the stable control of weld penetration depth under variable heat dissipation conditions.

Graphical abstract



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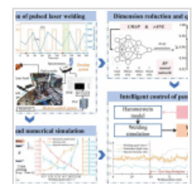

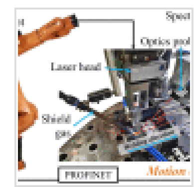

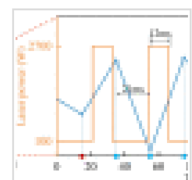

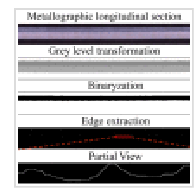

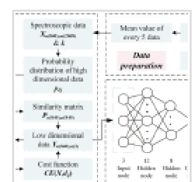

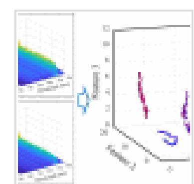

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Pulsed laser welding; Emission spectrum diagnosis; Feature extraction; Prediction of penetration depth; Model-free adaptive control (MFAC)

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1. Introduction

Pulsed laser welding has been widely used in the aerospace industry for joining titanium alloy components due to advantages of high energy density, low heat input and high welding efficiency [1]. Owing to the unique periodic nature of the laser power during the welding process, the weld penetration exhibits regular and complex fluctuations. In addition, the inconsistent heat-dissipation conditions in different joint areas of the titanium alloy components cause the fluctuation of the weld penetration, which further affects the mechanical properties of the joint [2], [3]. Therefore, online detection and real-time control of the penetration depth during the welding process is of great significance for ensuring the welding quality of the titanium alloy.

Precise online detection of the penetration depth is the basis for real-time control. However, current technological means cannot directly measure the penetration depth [4]. Instead, it is necessary to apply sensors to obtain acoustic [5], [6] electrical [7], and optical signals [8] that characterize the weld quality, thereby indirectly achieving online detection of the penetration depth. Passive electric probe is commonly used for collecting electrical signals in laser welding [9]. Nevertheless, the assembly requirements for such sensors are relatively strict, which makes it difficult for this collection method to meet the diverse production needs of different structural components. In addition, acoustic signals are easily interfered with by noise in complex working environments, which will affect the signal quality [10]. Regarding optical signal collection, the high-precision optical coherence tomography system has been applied in recent years, but the high cost makes it difficult to be widely promoted [11]. Spectrometer collects rich information from plasma optical radiation signals with high resolution and are widely used in welding process monitoring [12], [13]. It is worth mentioning that the spectral signal also has the problem of information redundancy which affects the detection accuracy [14], [15]. To this end, Yu et al. [16] used principal component analysis (PCA) linear dimensionality reduction to process the arc plasma spectral signal, approximating the original complex spectral data with a minimal amount of data. It is well known that the pulsed laser welding process is a complex nonlinear and unstable process. Therefore, the collected optical radiation signal does not have obvious regularity, which brings challenges to the data feature extraction. Zhao et al. [17] used different dimensionality reduction techniques to extract the features of dynamic resistance signals, established regression models for nugget diameter based on different feature inputs. They found that the regression model based on locally linear embedding (LLE) had the best performance, which proved the effectiveness of the automatic feature extraction algorithm based on nonlinear dimensionality reduction techniques. In 2018, the nonlinear dimensionality reduction technique uniform manifold approximation and projection (UMAP) was proposed [18], which is suitable for generating high-quality embeddings of large data sets and has been applied in engineering practice [19]. In this study, several dimensionality reduction techniques including UMAP are compared to extract plasma spectral features, and the best spectral features that can be used for real-time detection are explored.

The quality of welded joints directly affects the service life and structural reliability of products. Establishing closed-loop control strategies based on process signals has been widely recognized as an effective approach for ensuring weld formation quality [20], [21]. In recent years, advanced control methods, such as active disturbance rejection control (ADRC) and model-free adaptive control (MFAC), have demonstrated strong adaptability in manufacturing processes involving multi-physics coupling [22]. Previous studies have shown that MFAC can achieve stable and consistent penetration depth control in variable polarity plasma arc welding (VPPAW) [23]. Owing to its independence from accurate physical or mathematical models, MFAC exhibits strong robustness against parameter uncertainties and time-varying disturbances [24], [25], making it well suited for real-time penetration depth control in welding processes. However, compared with arc welding, pulsed laser welding involves intense plasma evolution and metal vapor recoil pressure, which lead to significantly more complex observation information and substantially increase the difficulty of precise intelligent control [26]. Therefore, it is urgent to develop and optimize control strategies specifically formulated for pulsed laser welding processes in order to ensure stable penetration regulation and reliable joint formation quality.

In summary, this study focuses on the core issue of forming quality control during pulsed laser welding. A simulation model of the temperature field for titanium alloy laser welding was constructed to systematically analyzes the keyhole evolution and plasma eruption mechanism during laser welding. Single-factor gradient welding experiments are designed to reveal the variation law of plasma spectral signals with respect to weld formation characteristics under varying welding speeds. Subsequently, multiple dimensionality reduction techniques are employed to extract representative spectral features, and a backpropagation neural network is developed for penetration depth prediction. Furthermore, several spectral intensity ratios are selected as the controlled variable, and a model-free adaptive controller for controlling the weld penetration is designed based Hammerstein model. The proposed strategy aims to achieve a balance between control stability and accuracy in online pulsed laser welding. By integrating time-domain feature smoothing with equivalent dynamic modeling, an effective representation of the complex process characteristics is realized, providing a new perspective for achieving high-quality laser welding.

2. Experimental procedure

The experimental platform for titanium alloy laser welding depth control is shown in Fig. 1, which is mainly composed of a pulsed laser welding system, a plasma spectrum sensing system and a motion control system. The laser welding system includes a laser generator, a laser head and a workpiece. According to the corporate process specifications for pulsed laser welding of titanium alloys, the laser pulse period is set to 33 ms. The spectral sensing system consists of a spectrometer, an optical cable and a fiber optics probe. The probe receives the spectral signal emitted by the plasma in real-time from the front side of the welding position and transmits it to the spectrometer through the cable. In addition, the acquisition frequency of a single spectral sample is set to 50 Hz, and a transmittance of 0.3% is equipped in front of the fiber optics probe so as to meet the limitation of the spectral intensity measurement range of the spectrometer. A motion control system is composed of Siemens PLC, robots, and an industrial computer. To minimize equipment synchronization delay, the industrial computer communicates with the PLC through the S7 protocol, which offers high reliability and efficient data transmission, enabling high-frequency real-time data updates with low network load. Simultaneously, the PLC and robots communicate via the PROFINET protocol that employs real-time communication technology (RT) and a periodic data transmission mechanism to complete data exchange within milliseconds. This efficient communication mode ensures timely transmission of control signals between the PLC and robots, thereby realizing closed-loop control between the industrial computer and robots.

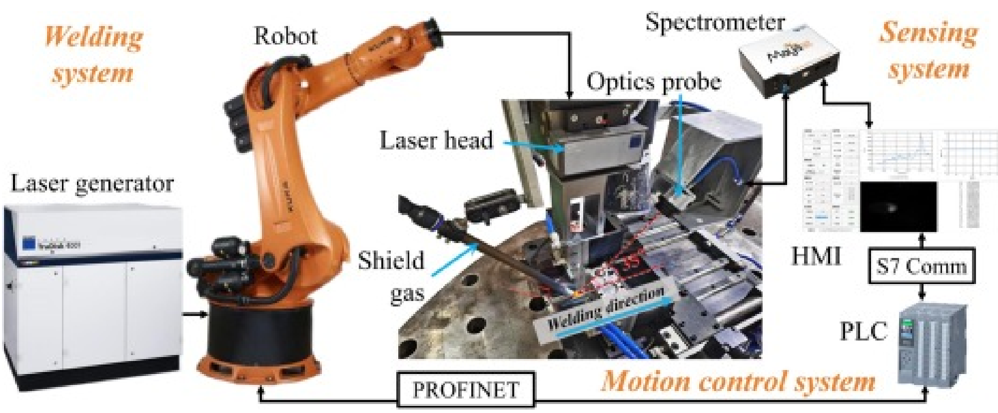

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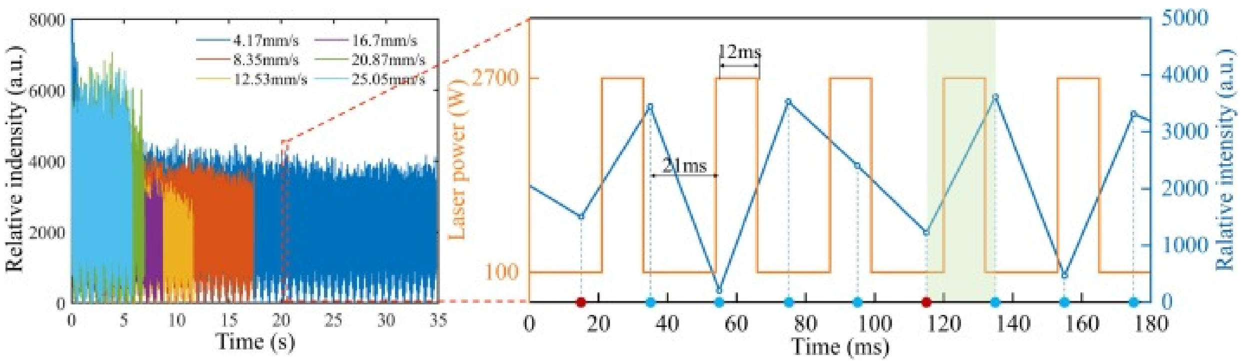
Fig. 1. Schematic of the pulsed laser welding experimental platform.

To further elucidate the intrinsic relationship between spectral features and penetration depth, this study established a transient three-dimensional thermal-fluid coupling numerical model for the laser pulse welding process of TC4 titanium alloy using the finite element analysis method and ANSYS® FLUENT software. The model strictly adhered to actual welding process parameters. The experiments and simulations were conducted on specimens with a thickness of 5 mm. A detailed description of the numerical simulation model construction is provided in the Supplementary Materials. The quantitative influence of welding speed on the spectral signals of the molten pool and weld formation quality was systematically examined through numerical simulation, providing a theoretical basis for online monitoring and control.

The spectral intensity data of Ti I 503.995 nm line collected at different welding speeds is shown in Fig. 2. It can be seen that the integration time for each of the five spectral sampling points is approximately equal to the time of three pulse periods. Therefore, by summing the five spectral sampling points, the total spectral intensity under the action of three laser pulses can be acquired, and the spectral intensity of a single pulse cycle can be calculated by the average value. After welding, the longitudinal section of the weld is processed as shown in Fig. 3 to extract the weld penetration data.



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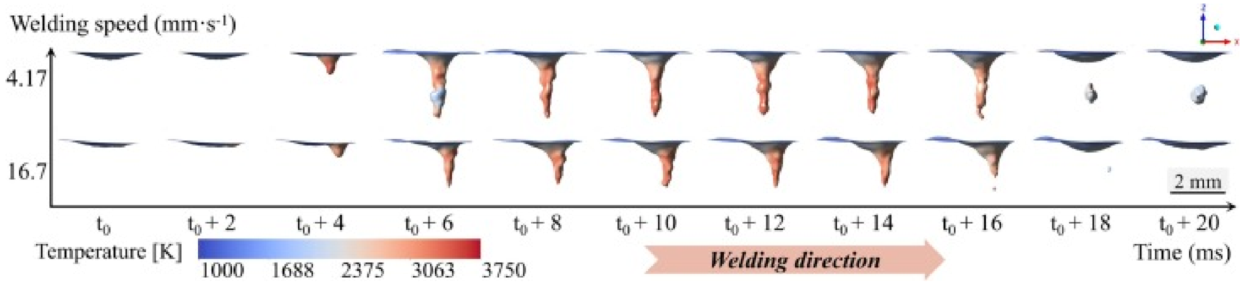

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Fig. 2. Spectral data acquisition and numerical simulation of pulsed laser welding process.

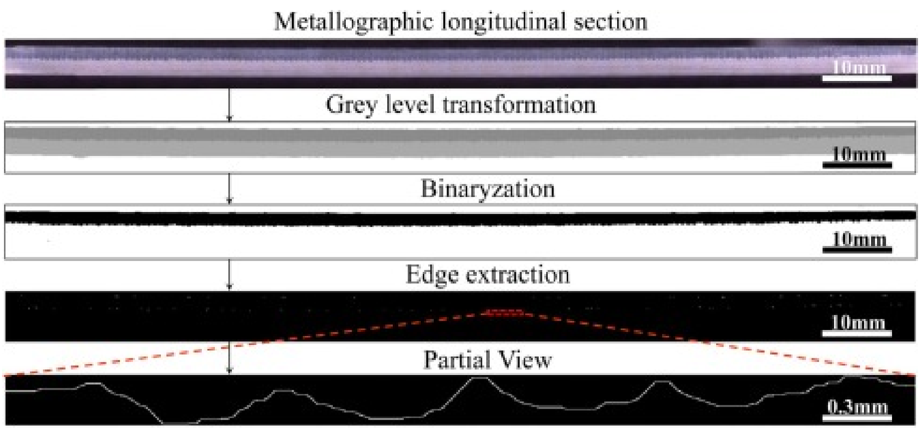

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Fig. 3. Illustrative view of weld penetration extraction.

In order to realize the online detection of the penetration depth and obtain the feedback signal required for penetration depth control, it is necessary to establish a relationship between the plasma spectral signal and the weld penetration depth by studying the change law of the plasma spectral signal and the weld penetration depth under different welding process parameters. For the sake of making the penetration depth within the acceptable range of engineering application, the test parameters are set and listed in Table 1.

Table 1. Parameters of pulsed laser welding.

Parameter	Value	Parameter	Value
Laser peak power (W)	2700	Laser base power (W)	100
Beam parameter product (mm·mrad)	4	Laser wavelength (nm)	1030
Workpiece (mm)	150 × 40	Shield gas	99.9%Ar
Dimension (mm)	× 5	Shield gas flow (L/min)	30
Laser pulse period (ms)	33	Pulse peak time (ms)	12

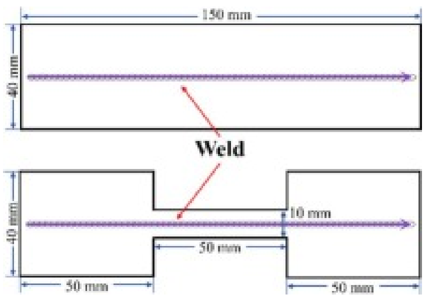
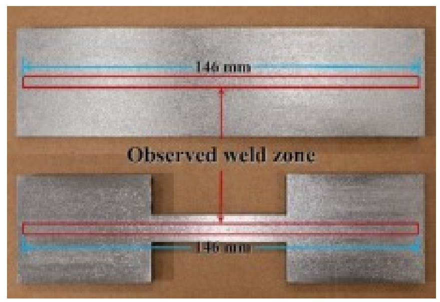
In previous studies, orthogonal experiments were designed with welding speed, laser power and pulsed period as the parameters. The experimental results show that laser power and welding speed are the main parameters that determine the weld penetration during pulsed laser welding. Since the adjustment permission of laser power cannot be obtained, the welding speed is selected as the control input in this study, and the variable welding speed experiment is designed. The constant speed experiment parameters are shown in Table 2, under the premise of ensuring good weld formation.

Table 2. Experimental parameters.

Experiment No.	Welding speed (mm/s)	Experiment No.	Welding speed (mm/s)
1	4.17	4	16.70
2	8.35	5	20.87
3	12.53	6	25.05

The change of heat-dissipation conditions is one of the most common working conditions in actual production, which will lead to variations in the welding temperature field and affect the welding quality. To verify the performance of MFAC by simulate the actual welding conditions, the welding process experiments designed in this study are shown in Table 3. Among them, Test-1 represents the welding conditions with constant heat-dissipation conditions, and Test-2 represents the welding conditions with sudden changes in heat-dissipation conditions.

Table 3. Experimental design of welding process.

Test No.	Model diagram	Workpiece object
1		
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3. Estimation of weld penetration

It is seen from Fig. 2(a) that the spectral intensity during pulsed welding process presents a nonlinear correlation with the welding speed. Results has shown that the emission intensity of a single spectral line during laser welding exhibits a nonlinear dependence on weld penetration depth. Initially the heat input decreases as the welding speed increases, leading to a reduction in the number of excited plasma particles. Consequently an apparent decrease in spectral intensity is observed. However, at higher welding speeds, the weld cross-sectional morphology changes significantly, leading to a pronounced increase in the depth-to-width ratio. This shift may be one of the reasons of the nonlinearity of intensity. The non-monotonic behavior caused by the complex interplay between heat input, plasma dynamics, and spectral emissions in laser welding processes will be analyzed in detail from the principle level in the subsequent part.

In summary, the collected original spectral intensity cannot be directly applied for detection models. At the same time, studies have verified that there is a strong correlation between the optical information from emitted in the molten pool area and the molten depth [27]. Therefore, t-distributed stochastic neighbor embedding (t-SNE) and UMAP feature reduction methods are used respectively to extract features from the spectral data to obtain two sets of plasma spectral features that characterize penetration depth in this section. Accordingly, detection models of penetration depth are established based on the input of different plasma spectral features, and the results are evaluated.

3.1. Dimensionality reduction

UMAP was proposed by McInnes [18] in 2018. Compared with t-SNE proposed, UMAP not only retains the global structure of the original data, but also retains the local structure better and has higher efficiency. The original spectral data set is comprised of 688 samples, each of which consists of 2068 wavelengths. UMAP is used to reduce the original spectral data $X_{n(688),m(2068)}=\{x_1, x_2, x_3, \dots, x_{688}\}^T$ to 3 dimensions. The probability distribution of high-dimensional data is expressed as follows:

$$p_{i|j} = e^{-\frac{d\left(x_i,x_j\right)-\rho_i}{\sigma_i}} \tag{1}$$

where $d(x_i,x_j)$ is the distance between two data points, σ_i is the variance of Gaussian distribution centered on data point x_i, and ρ_i is the distance between the i-th data point and the nearest data point. Compared with t-SNE, the formula of high-dimensional data point relationship removes the regularized term in the denominator, reducing the complexity of the calculation. Then, the value of σ_i is calculated by using the bisection method in (2). The similarity matrix $P_{n(688),n(688)}$ is formed using the joint probability distribution (3) for high-dimensional data.

$$k = 2^{\sum_i p_{ij}} \tag{2}$$

$$p_{ij} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i} \tag{3}$$

where k is the nearest neighbor, and its value was set before dimensionality reduction. The Laplace matrix is derived by the obtained $P_{n(688),n(688)}$. Calculate and take the eigenvectors that correspond to $a(3)$ eigenvalues with smaller values to form the initial array $Y_{n(688),a(3)}$ ={y_1, y_2, y_3, …, y_{688}} of the low-dimensional matrix. Similarly, the joint probability distribution of the low-dimensional data is calculated by (4) to form a similarity matrix $Q_{n(688),n(688)}$.

$$q_{ij} = \left(1 + a(y_i - y_j)^{2b}\right)^{-1} \tag{4}$$

The values of parameters a and b are determined by the fitting curve $\Psi(x, y)$ in (5).

$$\psi(x,y)=\begin{cases} 1 & if\|y_i - y_j\|^2 < min\_dist \\ \exp\left(-\|y_i - y_j\|^2 - min\_dist\right), & otherwise \end{cases} \tag{5}$$

The cost function of UMAP algorithm usually adopts binary cross-entropy (CE). The term that is only relevant to $p_{ij}(X)$ can be omitted as a constant term in the subsequent calculation of the gradient, in which case CE can be written as:

$$CE(X,d_{ij})= \sum_i[(1-P(X))\log(1-Q(d_{ij}))-P(X)\log Q(d_{ij})] \tag{6}$$

where d_{ij} represents the distance between y_i and y_j. The matrix $Y_{n(688),a(3)}$ is updated by stochastic gradient descent (SGD) in (7) until the gradient is less than the set threshold, and the optimal solution $Y_{n(688),a(3)}$ is obtained. $Y_{n(688),a(3)}$ is a set of data points with three features that are similar to the original data point distribution and have the same number of data points, retaining the information carried by the original data points in high-dimensional space. The process of UMAP algorithm is shown in Fig. 4.

$$\frac{\delta CE}{\delta y_i} = \sum_j\left[\frac{2abd_{ij}^{2(b-1)}P(X)}{1+ad_{ij}^{2b}} - \frac{2b(1-P(X))}{d_{ij}^2(1+ad_{ij}^{2b})}\right](y_i-y_j) \tag{7}$$

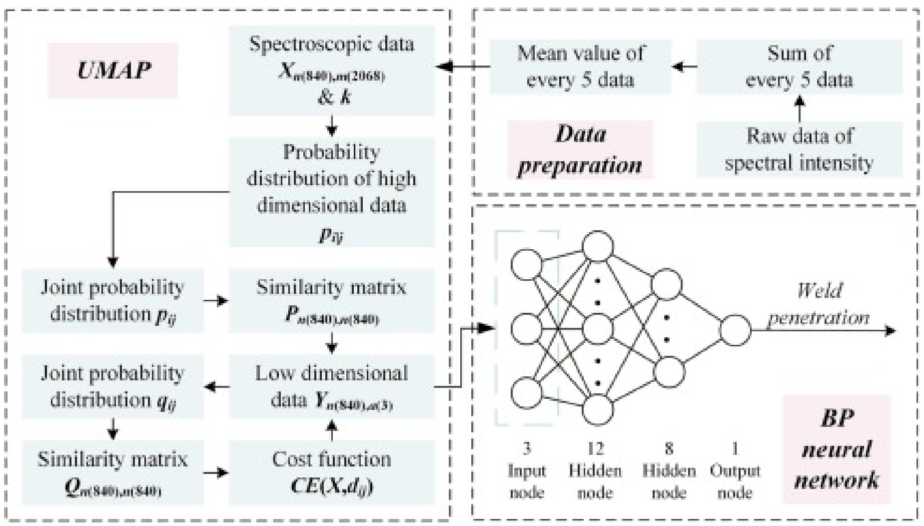

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Fig. 4. Structure of UMAP and pulsed laser welding penetration detection model.

3.2. Establishment of detection model

This subsection constructs a back propagation (BP) neural network with two hidden layers to predict the weld penetration of pulsed laser welding. The spectral signal features of the stable stage in the middle of welding are selected as the model input, the tansig function is selected as the activation function, and the corresponding weld penetration value is chosen as the model output. In this research, the structure of the detection model based on the laser-induced plasma spectral features is shown in Fig. 4. The BP neural network consists of four connection layers, namely an input layer with 3 input nodes, two hidden layers with 12 and 8 hidden nodes respectively and an output layer with only one output node. The accuracy of the model is evaluated using R-squared (R^2), Root Mean Square Error (RMSE) and Mean Absolute Error (MAE). The dataset was partitioned into training, validation and test subsets at a ratio of 7:1:2. The MAE on the validation set was monitored, and an early stopping strategy was employed. Model training was terminated when the validation MAE no longer decreased.

For all spectral feature configurations, training was terminated within 100 epochs according to the validation-based early stopping criterion. This behavior indicates stable convergence of the model parameters and suggests that the spectral data were sufficient to support effective model training. Fig. 5(a) shows spectral data at different welding speeds, and data processed by UMAP is visualized in Fig. 5(b). The discrimination of data is evident, indicating retention of high-dimensional data information. Fig. 6(a) presents UMAP prediction results, and Fig. 6(b) evaluates prediction models based on test set by t-SNE and UMAP. It is seen that processed data can accurately predict weld penetration. Both UMAP and t-SNE use probabilistic models to map high-dimensional data to low-dimensional space, preserving local neighborhood relationships. It is worth mentioning that UMAP follows the basic idea of manifold learning, which assumes that the data points are distributed in a low-dimensional manifold that is locally close to Euclidean geometric relations. UMAP attempts to maintain the global topology of the data, which allows it to map data points more efficiently while maintaining the overall relationship between clusters. The spectral data correlates strongly with welding speed, so extracting feature-process correlations to maintain global structure of dataset enhances dimensionality reduction quality. As is shown in Fig. 6(b), UMAP outperforms t-SNE, proving the effectiveness of balancing local and global data structures. The UMAP-based model offers superior prediction accuracy.

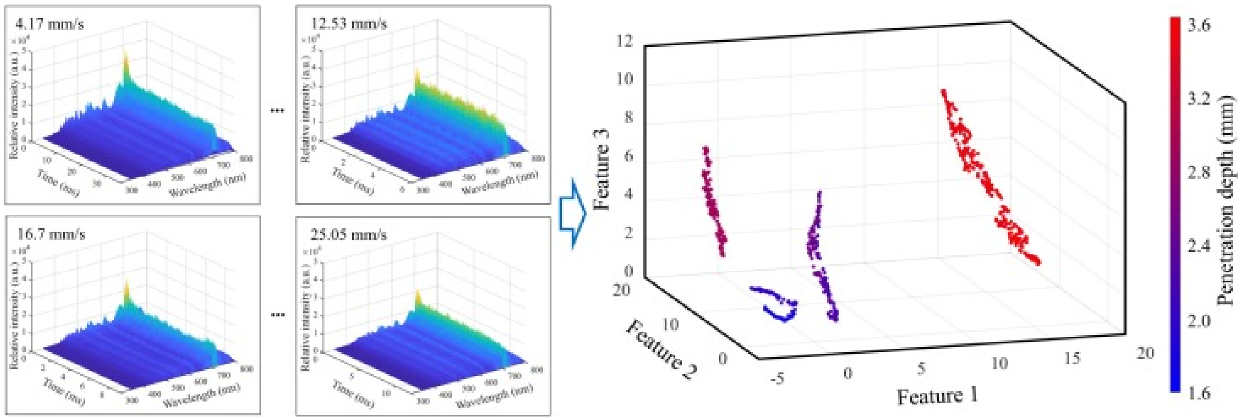
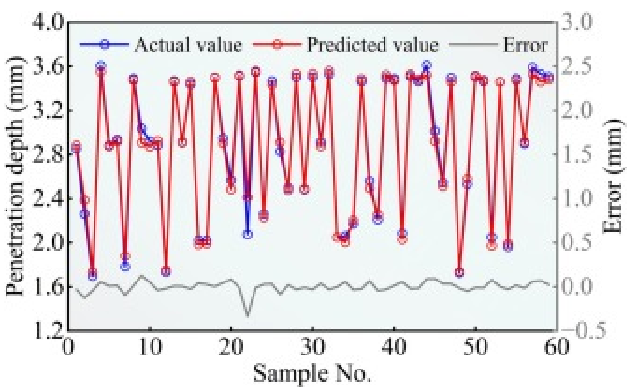


Fig. 5. Visualization process of UMAP based on collected spectral data



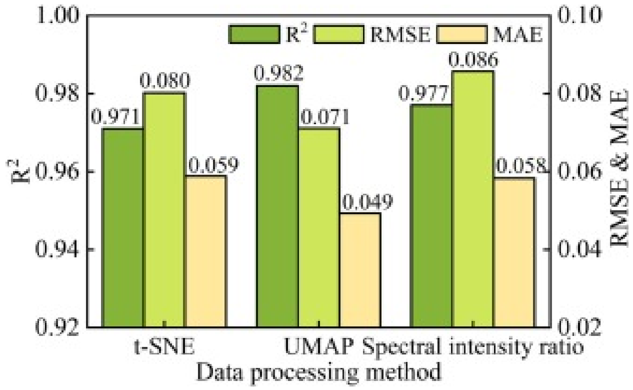


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Fig. 6. Model prediction results of different feature inputs.

The Spearman two-variable correlation analysis method is applied to analyze the performance of two dimensionality reduction methods. As shown in Fig. 7, data processed by UMAP generally gets higher coefficients, demonstrating that UMAP performs better extraction ability. It should be noted that UMAP and t-SNE were employed exclusively for offline feature exploration and validation, while the feedback variable used in the closed-loop control was directly derived from the averaged spectral data. In order to meet the needs of online detection of weld penetration during the laser welding process, a penetration detection model based on the plasma spectrum intensity ratio is established in the subsequent work.

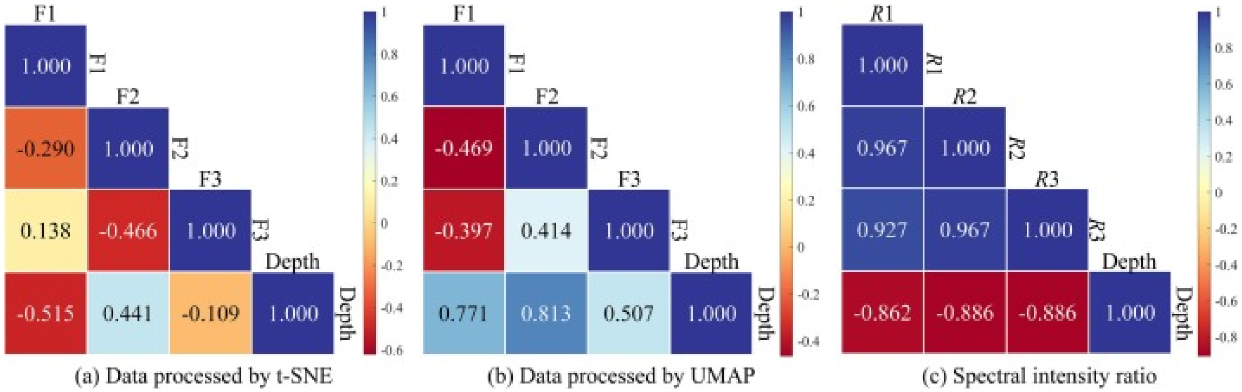

Fig. 7. Correlation coefficients of data processed by different methods.

4. Model-free adaptive control of weld penetration

Under different process parameters, the heat input during welding will affect the number of photons generated by the different electron energy level transitions of various elements, resulting in differences in the intensity of the different spectral lines. This paper selects 4 spectral lines with intense emission and calculates the ratios. Ti I 503.995 nm / V I 521.36 nm is denoted as $R1$. Ti I 503.995 nm / V I 551.72 nm is denoted as $R2$, and Ti I 503.995 nm / Ti I 586.919 nm is presented as $R3$. Emission lines denoted as Ti I and V I are selected for analysis. 'Ti I' and 'V I' represent neutral titanium and neutral vanadium atomic emission lines, respectively, where the Roman numeral 'I' indicates the neutral atomic state. These spectral lines originate from electronic transitions of metal atoms evaporated from the molten pool and excited in the laser-induced plasma. Neutral atomic lines were selected instead of ionic lines due to their stronger intensity and higher signal stability under the investigated welding conditions. As depicted in Fig. 7, the spectral line $R3$ with the highest absolute value of correlation coefficient is selected as the research object to analyze the changes in the average spectral intensity ratio under different process parameters. During the pulsed laser welding process, the penetration decreases monotonically with the increase of welding speed, and the specific plasma spectral intensity ratio increases monotonically with the increase of welding speed.

The average line diagram of weld penetration depth, *R*3 and spectral intensity of Ti I 503.995 nm at different welding speeds is shown in Fig. 8. Standard deviation error bars derived from repeated experiments were included to characterize the fluctuation amplitude of both the spectral signal and penetration depth data. The magnitude of the error bars reflects the intrinsic instability induced by thermo-fluid behavior and dynamic keyhole evolution during welding process. During the pulsed laser welding process, the penetration decreases monotonically with the increase of welding speed, and the corresponding error bars remained clearly separated across the investigated welding speed range. Likewise, the specific plasma spectral intensity ratio increases monotonically with the increase of welding speed. Although partial overlap was observed at 16.7 mm/s and 20.53 mm/s, the error bars of *R*3 remained largely separated across the six investigated welding speed conditions. The results show consistent trends of the spectral features and penetration depth with welding speed, suggesting basic predictability for subsequent data-driven modeling.

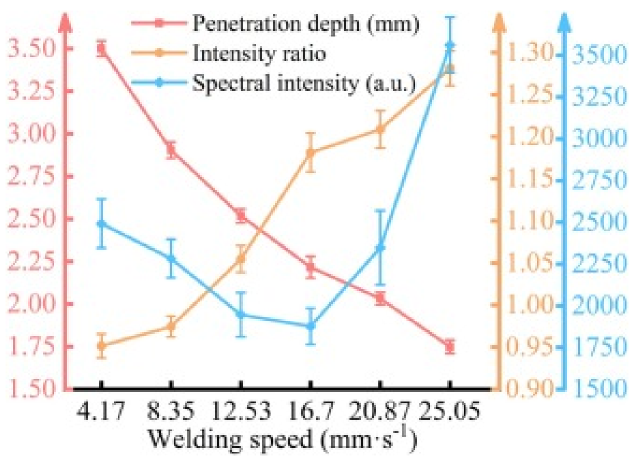

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Fig. 8. The change of penetration − spectral intensity ratio − spectral intensity with welding speed.

As shown in Fig. 9, although there is a nonlinear relationship between spectral intensity and penetration depth, different rates of change in welding speed may lead to monotonicity in the spectral intensity ratio. Specifically, while the intensities of both spectral lines first decrease and then increase, Ti I 586.919 nm has a lower energy level difference compared to Ti I 503.995 nm, making it more easily excited under the action of heat input. Therefore, the intensity ratio of these two spectral lines is relatively low at slow welding speeds. As the welding speed increases and the heat input decreases, the intensity of Ti I 586.919 nm decreases more significantly than that of Ti I 503.995 nm, causing the ratio to rise.

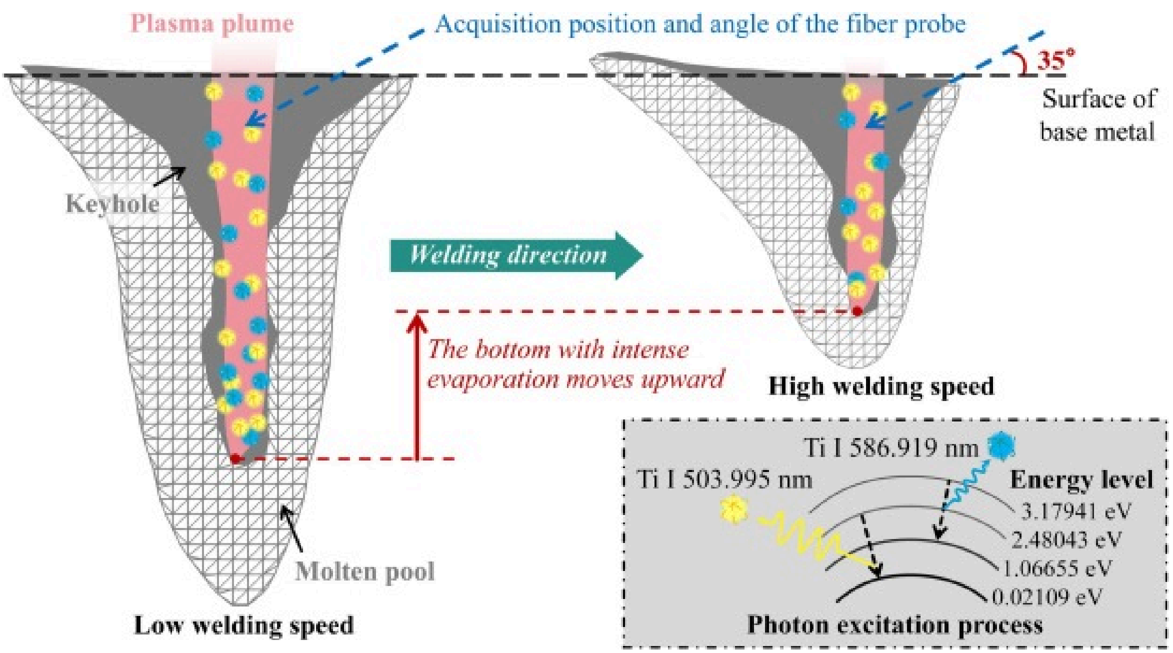

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Fig. 9. The variation trend of different spectral lines with the change of welding speed.

At high welding speeds, the factors influencing the intensity variations of Ti I emission lines become more complex. Reduced heat input under high-speed conditions leads to shallower weld penetration, causing the center of the plasma evaporation region to shift upward. As a result, the fiber optic probe collects signals from the region closer to the plasma core, that may cause nonlinear enhancement of the spectral signal. On the other hand, the reduced heat input significantly weakens the overall spectral intensity of the plasma plume. Under the combined influence of these two competing factors the intensity of a single emission line initially decreases with reduced thermal input, and subsequently exhibits uncertain trends. Further analysis of several strong Ti I emission lines reveals that both the Ti I 503.995 nm and Ti I 586.919 nm lines undergo a noticeable change in intensity trend at the same welding speed of 16.7 mm/s, demonstrating distinct sensitivity to process variation. As the welding speed exceeds 16.7 mm/s, the intensity of the 503.995 nm line transitions from a decreasing to an increasing trend, while the 586.919 nm line maintains a relatively stable intensity. This characteristic, where one spectral line increases while the other remains stable or continues to decrease, is essential for constructing an intensity ratio that exhibits a monotonic variation with welding speed. In contrast, other pairs of spectral lines fail to simultaneously meet this criterion, resulting in intensity ratios with poor monotonicity and limited discriminative capability. The spectral intensity ratio is influenced by multiple process parameters. In this work, parameters such as material thickness, shield gas flow and beam configuration were kept constant and not systematically explored.

It is worth mentioning that from Fig. 6(b) and Fig. 7, the correlation coefficient of $R3$ is significantly higher than that of the UMAP-reduced features, whereas its prediction performance is slightly inferior to that achieved by UMAP. Although the correlation coefficient between spectral feature data and penetration depth is used as an important indicator to measure the strength of the linear correlation, it does not necessarily correspond to the best predictive performance. Compared to other spectral intensity ratios, $R3$ demonstrates the most significant negative correlation with penetration depth and delivers the highest accuracy of prediction. It is seen in Fig. 6(b) that the detection model based on the spectral intensity ratio can realize online detection of the weld penetration, providing accurate information feedback for the subsequent closed-loop controller design. The original spectrum intensity ratio and weld penetration at different welding speeds is shown in Fig. 10. Therefore, in this section, the spectral line intensity ratio and penetration depths obtained at different welding speeds will be used to develop the MFAC algorithm for weld penetration control.

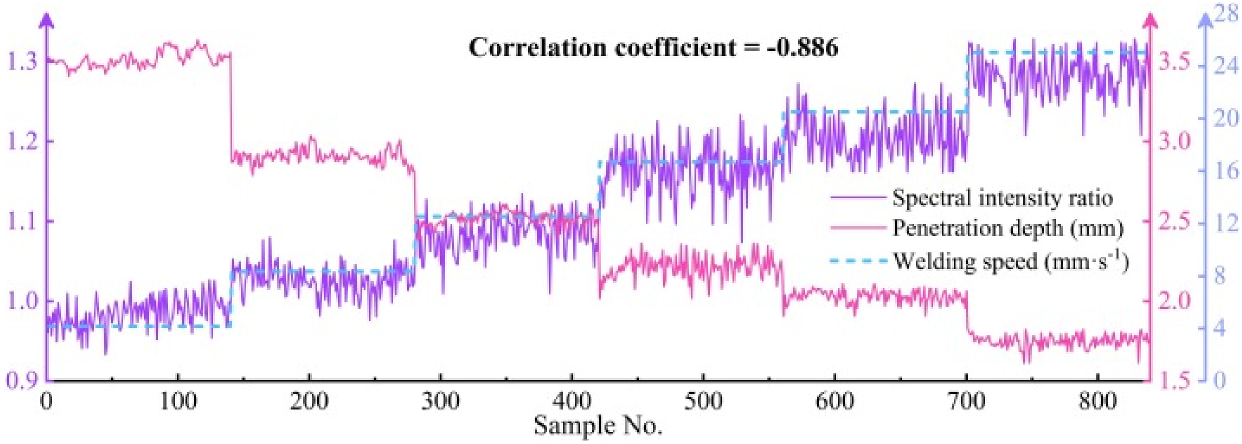

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Fig. 10. Weld penetration and spectrum intensity ratio at different welding speeds.

4.1. Establishment of the control model

4.1.1. Parameters optimization method

Currently, the methods for parameter identification of Hammerstein models mostly relies on bio-inspired intelligent optimization algorithms. Among the methods, Particle Swarm Optimization (PSO) has attracted considerable attention due to its simple structure, ease of implementation and wide applicability. Inspired by the social behavior of birds flocking or fish schooling in search of food, PSO simulates the cooperative and competitive behavior among individuals within a population to iteratively approach the global optimum in the search space.

In PSO, each particle represents a potential solution to the optimization problem. A particle adjusts its position in the multidimensional search space based on its own historical best position ***p****best* and the best-known position found by the entire swarm ***g****best*. Through iterative updates of velocity and position, particles tend to converge toward optimal regions.

Let the search space be $\boldsymbol{D}$-dimensional. The position and velocity of the i-th particle are denoted as:

$$x_i = (x_{i1}, x_{i2}, x_{i3}, \ldots, x_{iD}), v_i = (v_{i1}, v_{i2}, v_{i3}, \ldots, v_{iD}), \tag{8}$$

The personal best position is $\boldsymbol{p}_{\mathrm{i}}$, and the global best position is $\boldsymbol{g}$. The update rules for velocity and position are as follows:

$$v_{id}^{k+1} = \omega v_{id}^{k+1} + c_1 r_1 (p_{id} - x_{id}^k) + c_2 r_2 (g_d - x_{id}^k) \tag{9}$$

$$x_{id}^{k+1} = x_{id}^k + v_{id}^{k+1} \tag{10}$$

where k denotes the current iteration. ω is the inertia weight controlling the trade-off between global exploration and local exploitation. c_1 and c_2 are cognitive and social acceleration coefficients, respectively. r_1 and r_2 are random values drawn from a uniform distribution in [1]. The inertia component $\omega\, v_{id}^k$ maintains the particle's previous motion trend. The cognitive component $c_1 r_1(\boldsymbol{p}_{\mathrm{id}} - x_{id}^k)$ encourages the particle to return to its personal best position, and the social component $c_2 r_2(\boldsymbol{g}_{\mathrm{d}} - x_{id}^k)$ attracts the particle toward the global best position. The combination of these three elements guides the particle to new positions, thereby continuously improving the solution quality. Specific steps are as follows

Initialize the positions and velocities of all particles randomly within the search space. Set the population size N, maximum number of iterations

k_{max}, acceleration coefficients c_1 and c_2, inertia weight ω and velocity limits $[v_{min}, v_{max}]$. Evaluate the fitness of each particle and initialize personal and global bests.

(2) Compute the fitness value of each particle using the objective function, which is defined according to the specific optimization problem.

(3) For each particle, if the current position yields better fitness than its personal best, update the personal best. Update the global best with the best solution found by any particle in the population.

(4) Use the update equations to modify each particle's velocity and position. Clamp velocities to their allowable ranges to prevent divergence.

(5) Check whether the maximum number of iterations k_{max} has been reached or the set convergence criterion is satisfied. If not, return to Step (2). Otherwise terminate the process and output the best solution found. The standard PSO algorithm operates according to the iterative procedure illustrated in Fig. 11.

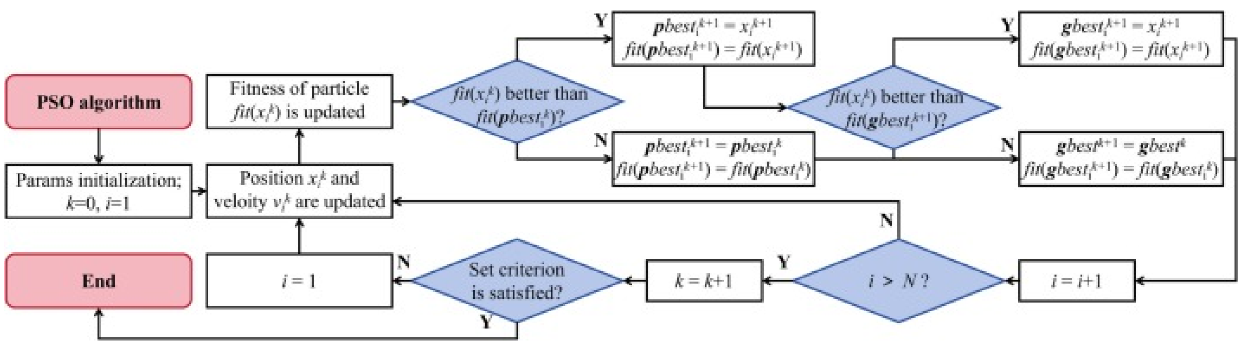

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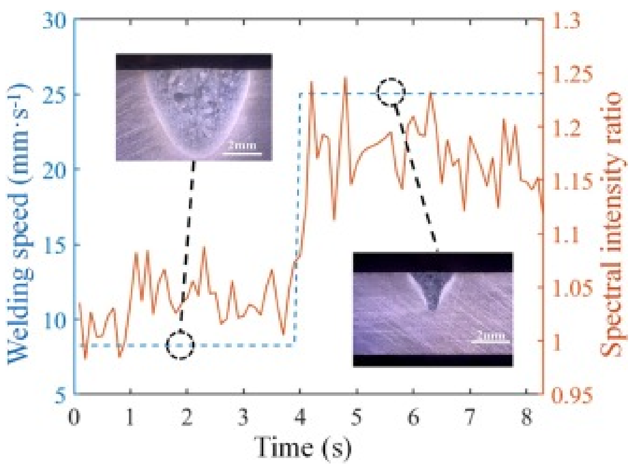
Fig. 11. Schematic of the PSO algorithm process.

4.1.2. Design of the model-free adaptive controller

In order to study the dynamic characteristics of the spectral signal features during laser welding, a positive and negative step experiment is designed. In industry, control systems are often designed and adjusted based on first-order inertial element [28]. For the welding process with laser as the heat source, this idea can be used to understand the dynamic characteristics of the spectral characteristic signals of the laser welding process. The time-domain response of the plasma spectral features is shown in Fig. 12, which is described by first-order inertia element with time delay.



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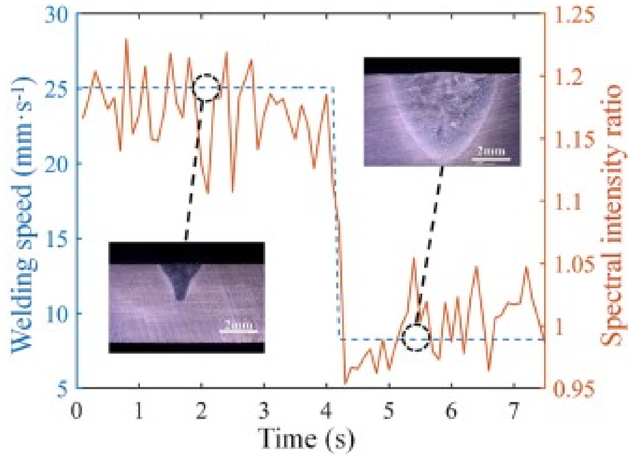

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Fig. 12. Time domain response of the plasma spectral characteristics.

It is considered that the laser welding process is a nonlinear and complex process, which is difficult to be accurately reflected by a simple linear model. In this study, the Hammerstein model of nonlinear systems is used to describe the relationship between the welding speed (u) and the plasma spectral intensity ratio (y) in the laser welding process as follows:

$$\begin{gathered} y(k) = -a_1 y(k-1) - a_2 y(k-2) - \cdots - a_{n_a} y(k-n_a) \\ +(b_0\gamma_0 + b_1\gamma_0 + \cdots + b_{n_b}\gamma_0) \\ +b_0\gamma_1 u(k-d) + b_1\gamma_1 u(k-d-1) + \cdots + b_{n_b}\gamma_1 u(k-d-n_b) \\ +b_0\gamma_2 u^2(k-d) + b_1\gamma_2 u^2(k-d-1) + \cdots + b_{n_b}\gamma_2 u^2(k-d-n_b) \\ +\cdots \\ +b_0\gamma_{n_\gamma} u^{n_\gamma}(k-d) + b_1\gamma_{n_\gamma} u^{n_\gamma}(k-d-1) + \cdots + b_{n_b}\gamma_{n_\gamma} u^{n_\gamma}(k-d-n_b) \end{gathered} \tag{11}$$

where $y(k) \in \mathrm{R}$ is the system output at the k moment, and $u(k) \in R$ is the system input at k. n_a and n_b determine the number of historical spectral intensity ratio measurements and welding speed variations involved in the control law. n_γ denotes the adaptive gain that determines the magnitude of welding speed adjustment in response to deviations in the spectral intensity ratio. d is a regularization parameter introduced to prevent excessive speed adjustments caused by minor or random spectral variations. This formula is written as follows as well:

$$y(k)= \varphi_1(k)\theta_1^T + \varphi_2(k)\theta_2^T \tag{12}$$

$$\varphi_1(k)=[-y(k-1), -y(k-2), \cdots, -y(k-n_a)]\in R^{1\times n_a} \tag{13}$$

$$\begin{aligned}\varphi_2(k)=&[1, u(k-d), u(k-d-1)\cdots, u(k-d-n_b),\\ &u^2(k-d), u^2(k-d-1)\cdots, u^2(k-d-n_b),\\ &\cdots,\\ &u^{n_\gamma}(k-d), u^{n_\gamma}(k-d-1)\cdots, u^{n_\gamma}(k-d-n_b)]\\ &\in R^{1\times[1+n_\gamma\times(n_b+1)]}\end{aligned} \tag{14}$$

$$\theta_1 =[a_1, a_1, \cdots, a_{n_a}]\in R^{1\times n_a} \tag{15}$$

$$\begin{aligned}\theta_2 =&[b_0\gamma_0 + b_1\gamma_0 + \cdots + b_{n_b}\gamma_0, b_0\gamma_1, b_1\gamma_1, \cdots, b_{n_b}\gamma_1,\\ &b_0\gamma_2, b_1\gamma_2, \cdots, b_{n_b}\gamma_2, \cdots b_0\gamma_{n_\gamma}, b_1\gamma_{n_\gamma}, \cdots, b_{n_b}\gamma_{n_\gamma}]\\ &\in R^{1\times[1+n_\gamma\times(n_b+1)]}\end{aligned} \tag{16}$$

Based on (12)-(16), the structural parameters n_a, n_b, n_γ, d and the parameter vector in the model is optimized by PSO. Mean squared error is set as the fitness function:

$$J = \frac{1}{L}\sum_{t=1}^{L}\left(y(k)-\left[\varphi_1(k)\theta_1(k)^T + \varphi_2(k)\theta_2(k)^T\right]\right)^2 \tag{17}$$

When n_a = 3, n_b = 3, n_γ = 2 and d = 1, the optimal solution is obtained for the parameters. At this time, the dynamic relationship model between the welding speed and the plasma spectral intensity ratio is:

$$\begin{gathered} y(k)=1.4524y(k-1)-1.2711y(k-2)+0.2307y(k-3) \\ -0.0002668+0.000070518u(k-1)+0.0007229u(k-2) \\ -0.000763u(k-3)+0.00032386u(k-4) \\ -0.00015384u^2(k-1)+0.00021758u^2(k-2) \\ +0.000014546u^2(k-3)-0.000065603u^2(k-4) \end{gathered} \tag{18}$$

When $\Delta u(k)=u(k)-u(k-1)\neq 0$, a Pseudo Partial Derivative (PPD) must exist that satisfies the following equation:

$$\Delta y(k+1)=\phi^T(k)\Delta u(k) \tag{19}$$

For the rule learning control algorithm, consider the following control input criterion function:

$$J(u(k))=[y(k+1)-y_r(k+1)]^2+\lambda\|\Delta u(k)\|^2 \tag{20}$$

where $y_r(k+1)$ is the expected output, and $y(k+1)$ is the actual output. λ is the weight constant, $\lambda>0$. Since $\phi(k)$ is unknown, the criterion function for its estimated value $\hat{\phi}(k)$ is:

$$J(\phi(k))=\left[y(k)-y(k-1)-\phi^T\Delta u(k-1)\right]^2+\mu\left\|\phi(k)-\hat{\phi}(k-1)\right\|^2 \tag{21}$$

where λ and μ are parameter weight factors, $\lambda>0$, $\mu>0$. According to (11) and (12), the core calculation formula of the MFAC is obtained as follows:

$$\hat{\phi}(k)=\begin{cases}\hat{\phi}(1) & if\ k=1\ or\ \hat{\phi}(k)\leqslant\varepsilon\ or\ sign\left(\hat{\phi}(k)\right)\neq sign\left(\hat{\phi}(1)\right)\\ \hat{\phi}(k-1)+\frac{\eta\Delta u(k-1)}{\mu+\|\Delta u(k)\|^2}\left[\Delta y(k)-\hat{\phi}^T(k-1)\Delta u(k-1)\right] & otherwise\end{cases} \tag{22}$$

$$u(k)=u(k-1)+\frac{\rho_k\phi(k)}{\lambda+\|\phi(k)\|^2}\left[y_r(k+1)-y(k)\right] \tag{23}$$

where $\hat{\phi}(1)$ is the initial value of $\hat{\phi}(k)$, ε is a very small positive number, which is taken here as 10^{-5}, ρ and η are the step size factors, usually $\rho\in(0,1]$, $\eta\in(0,1]$.

4.2. Control simulation of the welding process

For the pulsed laser welding process, the closed-loop control structure of penetration depth based on MFAC is shown in Fig. 13, and the specific implementation steps of the MFAC algorithm applied to the single-input single-output (SISO) system are shown in Table 4. Before the developed controller put into the actual process, simulation tests should be performed to verify its performance. The input of the simulation is the welding speed, and the output is the spectral intensity ratio. The desired spectral intensity ratio is $IRset=1.15$. The simulation parameters determined by the particle swarm optimization algorithm are $\eta=0.8262$, $\mu=0.00001$, $\rho=1$ and $\lambda=0.0049$. The simulation results of MFAC is showed in Fig. 14. It is seen that the algorithm basically meet the requirements of the pulsed welding process for weld penetration control, with a maximum overshoot of 4.3%.

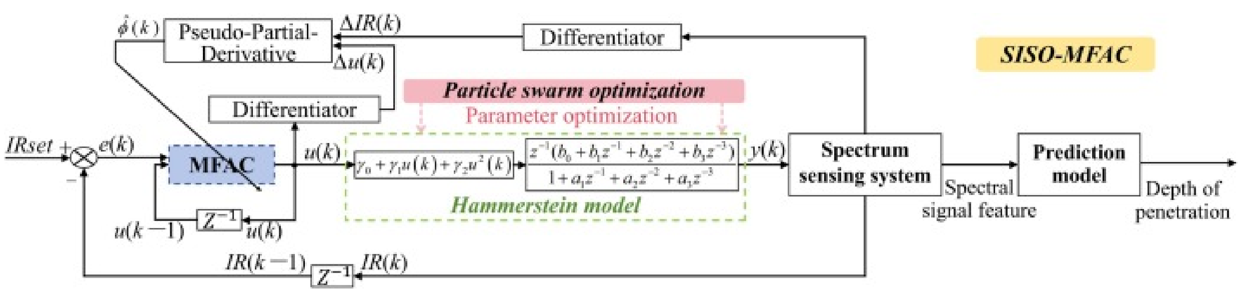

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Fig. 13. Pulsed laser welding single-input single-output MFAC system.

Table 4. MFAC methodology for penetration control.

Specific implementation step of MFAC algorithm:
Step 1: Calculate the estimated PPD based on the known input, output data $\{u(k), IR(k)\}$ and (13). Step 2: Obtain the error $e(k)$ at the current time in real-time by setting the expected spectral intensity ratio *IRset*, and the welding speed $u(k)$ to be adjusted at k moment is calculated based on the MFAC algorithm (14). Step 3: Apply the obtained welding speed $u(k)$ to the current pulsed laser welding process to obtain the I/O data set $\{u(k), IR(k)\}$ at the time of $k+1$; Step 4: Cycle through steps 1 to 3 to obtain a series of new data set $\{u(k), IR(k)\}$, $k=1, 2, \ldots$, until the end of the simulation time.

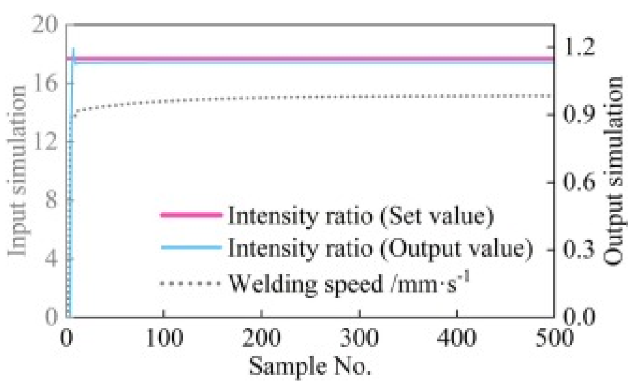

Fig. 14. Simulation curve of single variable MFAC.

5. Welding control experiments

5.1. Welding experiment with constant parameters

In order to demonstrate the adverse effects of changes in heat-dissipation conditions during the welding of titanium alloy materials and to provide a reference for controlling the experimental results, constant and variable heat-dissipation conditions are used for the constant specification experiments. Rectangular and dumbbell-shaped workpieces are used in the experiment. The specific welding process parameters are determined through experiments, and welding experiments need to be carried out under conditions that are consistent with the modeling conditions. When the weld quality meets the conditions, the welding speed is 16.7 mm/s.

The length of the positive weld seam for the constant-parameters process is 146 mm. The results of the constant-specification process are shown in Fig. 15. As can be seen from figures, the spectral line intensity ratio changes relatively with the change in heat-dissipation conditions. At the constant heat-dissipation condition, the spectral line intensity ratio tends to be consistent. At the position where the heat-dissipation condition is worse, the intensity ratio of plasma spectral line decreases, which reflects the increase of welding depth. This is because when the heat-dissipation conditions worsen, the temperature of the base metal rises, which on the one hand increases the temperature of the area around the weld pool and the size of the weld pool, resulting in an increase in the penetration, and on the other hand increases the evaporation of the metal and the recoil pressure, resulting in an increase in the depth of the keyhole and an increase in the penetration of the weld. Therefore, by adjusting the welding speed in real-time to control the amount of heat input to the weld, the penetration of the weld can be stably controlled under varying heat-dissipation conditions, as shown in Fig. 16.

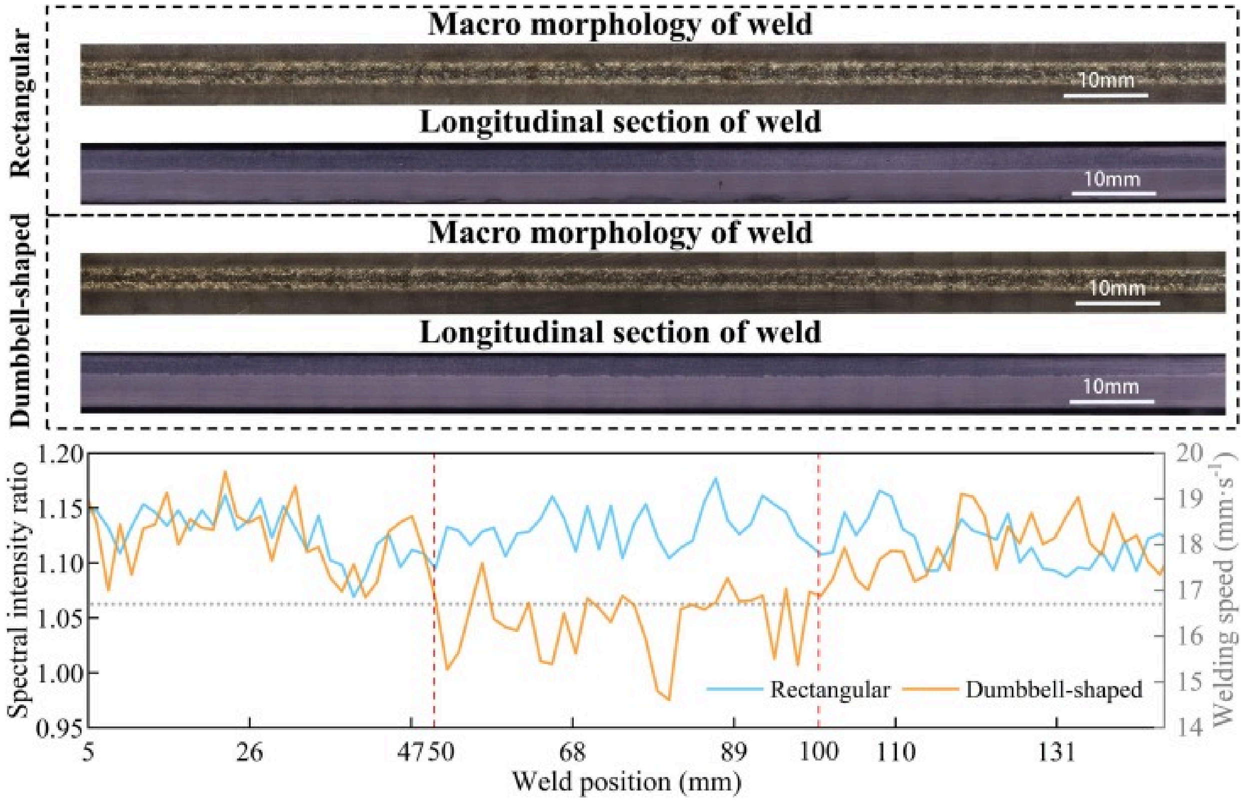

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Fig. 15. Result of welding experiment with constant parameters.

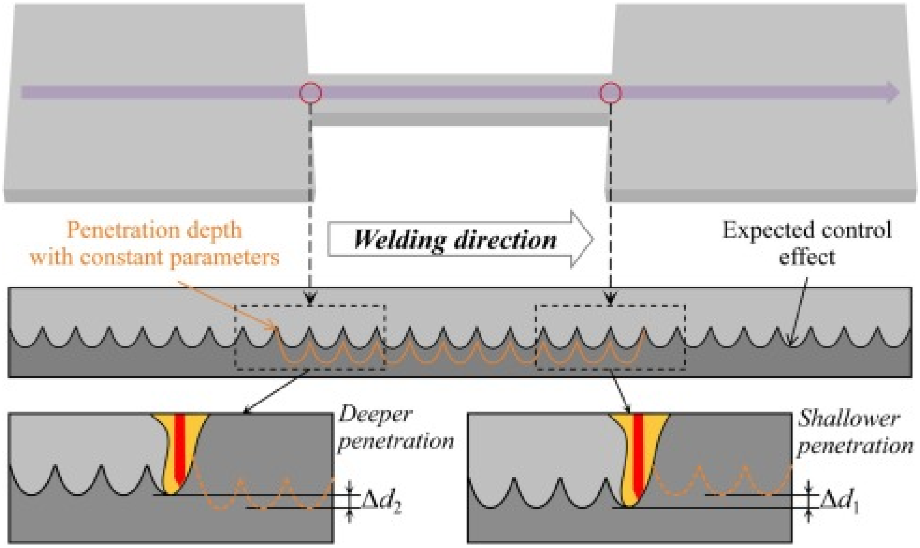

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Fig. 16. Illustration of control effect of MFAC.

5.2. Welding experiment with model-free adaptive controller

Based on the established communication, during laser welding, the spectrometer transmits the plasma spectral signals collected above the molten pool to the industrial computer. The computer converts the spectral characteristic signals into the difference values of control variables, and then transmits them in the form of electrical signals to the PLC through the Ethernet interface. This causes the corresponding parameters of the PLC to change, thereby controlling the robot to adjust the welding speed and achieving closed-loop control. The control cycle is 150 ms, which means the welding speed is adjusted once every 5 pulse cycles. In order to verify the closed-loop control effect of the penetration system, a control experiment is designed for a dumbbell-shaped workpiece named Test-2 with a sudden change in heat dissipation conditions. The system input is the welding speed, and the expected spectral intensity ratio is 1.15, corresponding to the weld penetration of 2.20 mm.

During the experiment, the welding speed is adjusted to control the weld penetration, which is expressed as the change in the intensity ratio of the plasma spectrum. Fig. 17 shows the results of the MFAC intelligent closed-loop control of the dumbbell-shaped heat-dissipation workpiece. The length of the weld is 146 mm, and the data of the beginning and the end of the welding process are removed. It is observed that the surface formation and penetration depth of weld on the dumbbell-shaped workpiece are relatively uniform when the MFAC algorithm is adopted. 87.3% penetration depth of a single weld is maintained at 2.20 ± 0.15 mm, with a standard deviation of 0.0986. Stable control of the weld penetration under variable heat-dissipation conditions is basically achieved. Although the proposed approach demonstrates promising performance under laboratory conditions, several challenges remain for industrial deployment. These challenges mainly relate to robustness against process disturbances, long-term reliability and compliance with industrial safety. Addressing these aspects will constitute an important direction for future work toward large-scale industrial application.

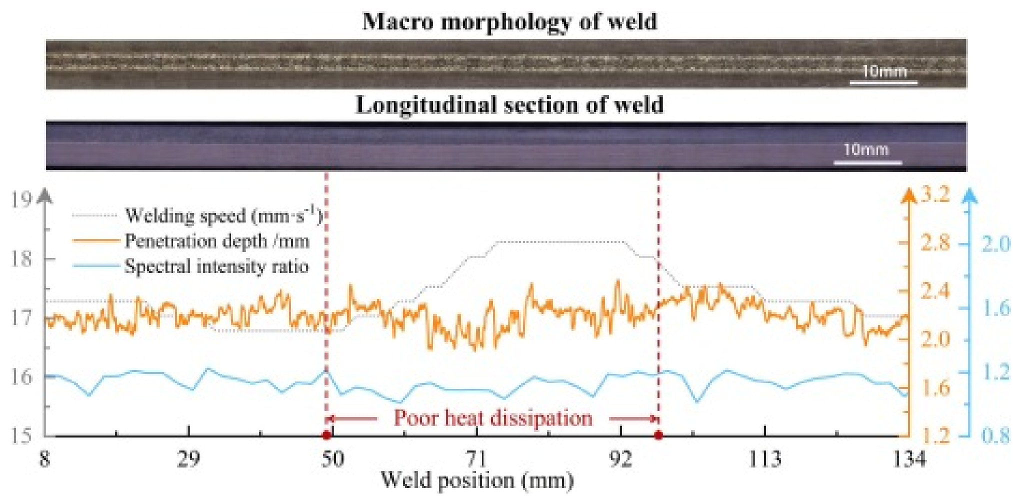

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Fig. 17. Control results of dumbbell-shaped workpiece using MFAC intelligent controller.

6. Conclusions

This paper analyzes the variation of plasma spectral features with welding process parameters and its relationship with penetration during pulsed laser welding, and then weld penetration detection models based on plasma spectral features are established. Furthermore, this study proposes the spectral intensity ratio of Ti I 503.995 nm / Ti I 586.919 nm that can represent the depth of penetration in real-time, and designs a model-free adaptive control system for plasma spectral signal monitoring and penetration control. The main conclusions are as follows:

The spectral intensity of the plasma exhibits a nonlinear relationship with the penetration depth, and cannot be applied directly for penetration depth detection. The spectral features processed by t-SNE and UMAP can be used to construct a penetration depth prediction model, and the model based on features by UMAP has higher accuracy.

The influencing factors of spectral intensity of Ti I lines varying with welding speed are elucidated, and the potential of intensity ratios for predicting weld penetration is comparatively evaluated. The spectral intensity ratio of Ti I 503.995 nm / Ti I 586.919 nm has a strong negative correlation with the penetration depth, with a correlation coefficient of −0.886, and can be used to represent the penetration depth in real-time.

The nonlinear characteristics of the dynamic output response of the plasma spectral characteristics are demonstrated by positive and negative step experiments, and the Ti I 503.995 nm / Ti I 586.919 nm spectral intensity ratio is available for controlling the penetration depth.

A pulsed laser welding depth control system is developed using model-free adaptive control algorithm with the welding speed as the control variable and the plasma spectral intensity ratio as the controlled variable. Simulation analysis results show that the intelligent controller has the characteristics of small overshoot. Control experiment on dumbbell-shaped workpiece shows that the designed control system can achieve stable control of the penetration depth under variable heat-dissipation conditions.

CRediT authorship contribution statement

Mingyu Li: Writing – original draft, Formal analysis, Data curation. **Kaisong Yang:** Writing – original draft, Methodology, Formal analysis. **Yunfu Tian:** Software, Formal analysis. **Yiming Huang:** Writing – review & editing, Funding acquisition, Formal analysis. **Lijun Yang:** Supervision, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following are the Supplementary data to this article:

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Supplementary Data 1.

Data availability

Data will be made available on request.

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